

Schematic V1.0

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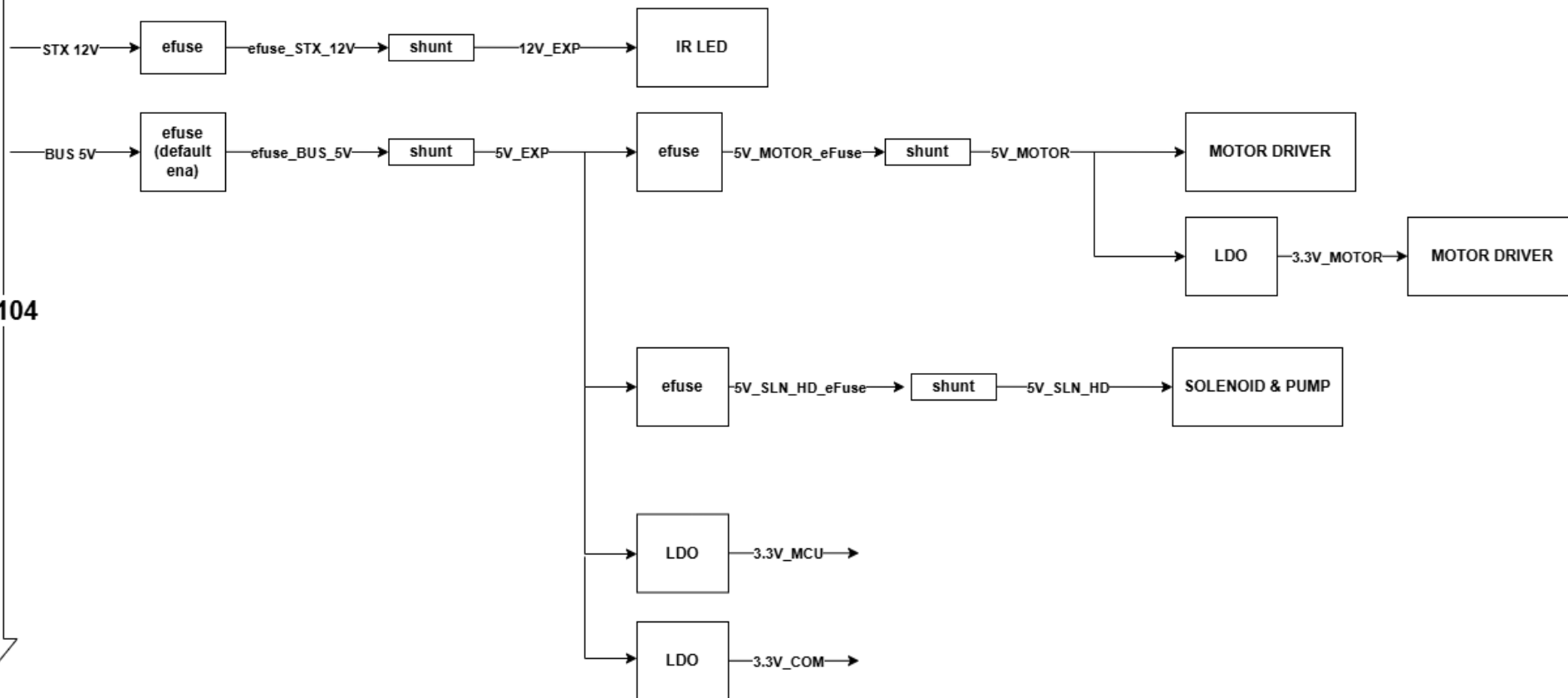
History

VERSION	DISCRIPTION
1.0	INITIAL PROJECT

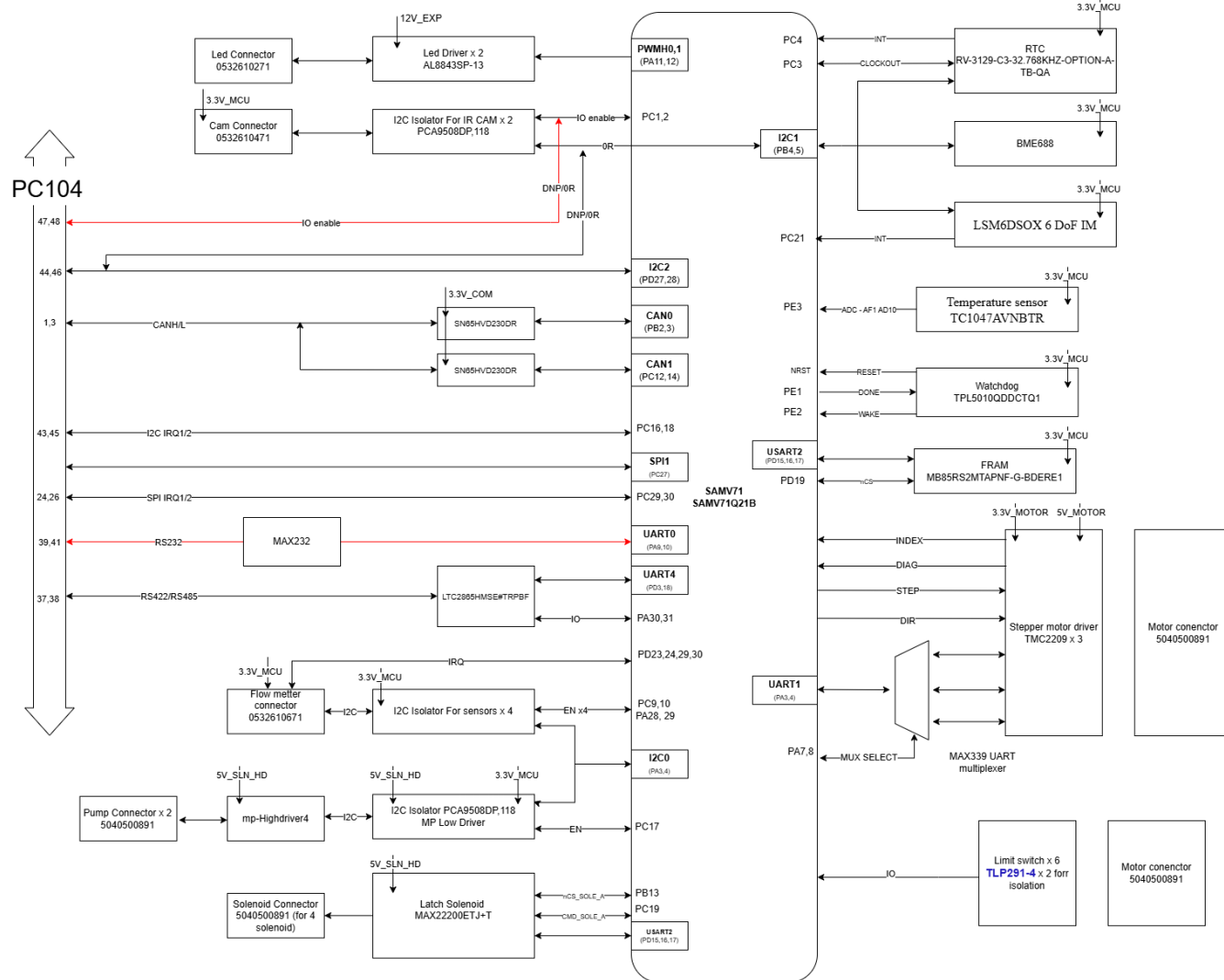


Project	BEE_DAL	Sub-system: BEE-DAL_EXP_V1.0.PrjPcb		
Design by:	Duy Nguyen	Sheet title: 01_INDEX.SchDoc	Sheet: 1	of 27
Checked by:	Bao Bui	Doc no: 1	Size: A3	
Approved by:	Vu Pham	Revision 1.0.0	Date: 11-Aug-26	

PC104

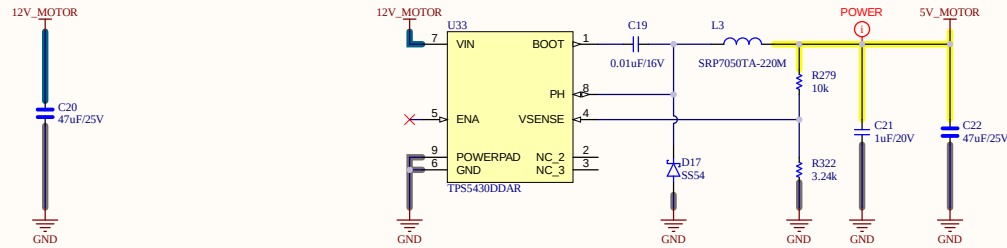


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Design by:	Duy Nguyen	Sheet title: 02_POWER_TREE.SchDoc	Sheet: 2 of 27
Checked by:	Bao Bui	Doc no: 2	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 11-Aug-26

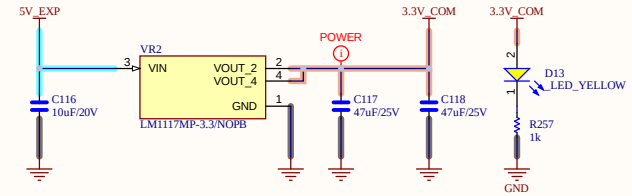


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Design by:	Duy Nguyen	Sheet title: 03_BLOCK_DIAGRAM.SchDoc	Sheet: 3 of 27
Checked by:	Bao Bui	Doc no: 3	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 11-Aug-26

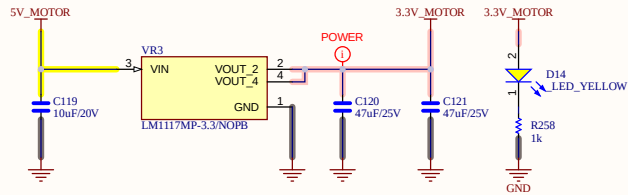
BUCK for PERIPHERAL



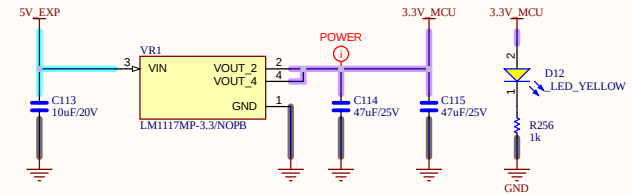
LDO for driver (driver communication)



LDO for motor



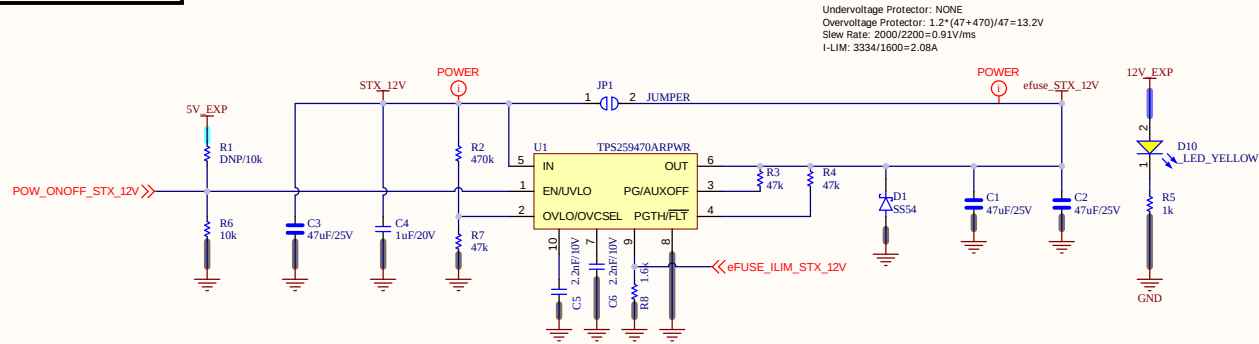
LDO for MCU



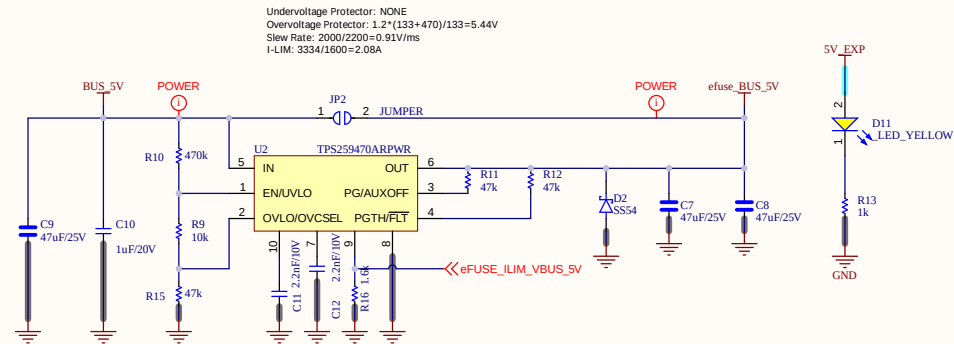
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Design by:	Duy Nguyen	Sheet title:	04_POWER.SchDoc
Checked by:	Bao Bui	Doc no:	4
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	4 of 27
		Size:	A3
		Date:	11-Aug-26

* 2 efuse dùng cho nguồn tổng là STX 12V và BUS 5V, trong đó STX 12V sẽ có thể được MCU on/off, còn BUS 5V thì mặc định on (do MCU lấy 5V từ đây)

EFUSE STX 12V

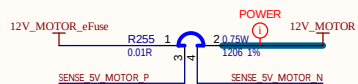
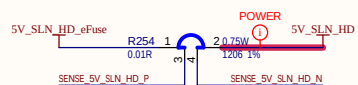
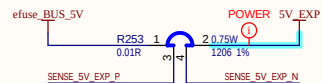
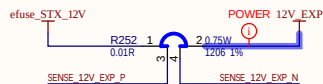
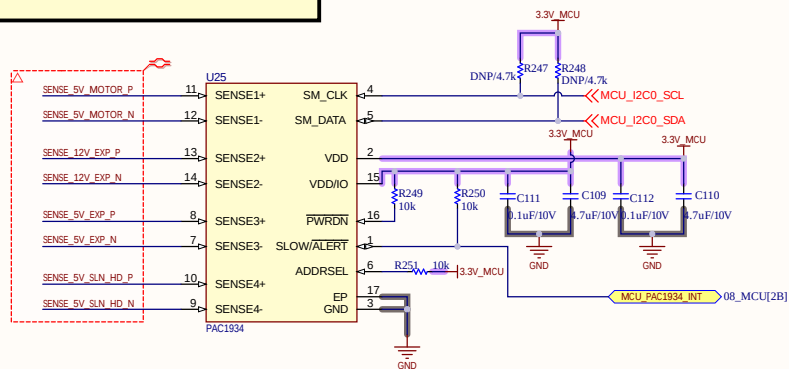


EFUSE BUS 5V



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title: 05_POWER_EFUSE.SchDoc	Sheet: 5 of 27
Checked by:	Bao Bui	Doc no: 5	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 11-Aug-26

POWER MEASUREMENT 12V AND 5V



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	06_POWER_MEASUREMENT.SchDoc
Checked by:	Bao Bui	Doc no:	6
Approved by:	Vu Pham	Revision	1.0.0
		Size:	A3
		Date:	11-Aug-26

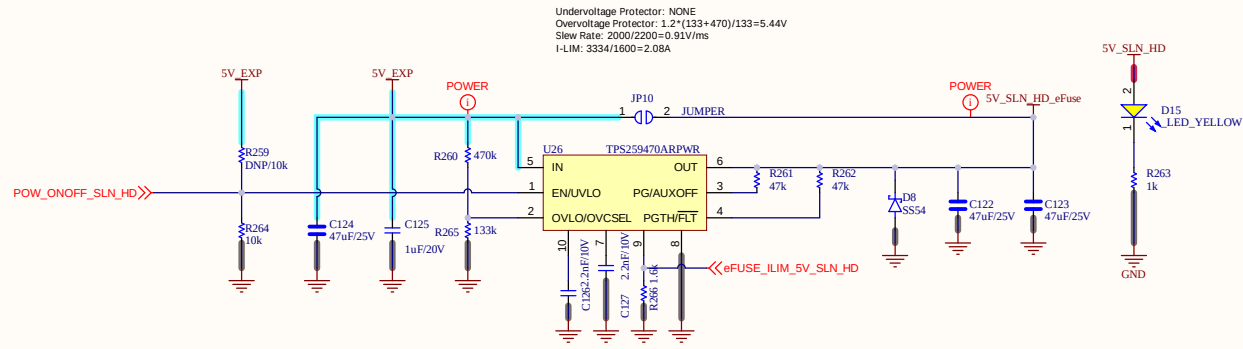
A

B

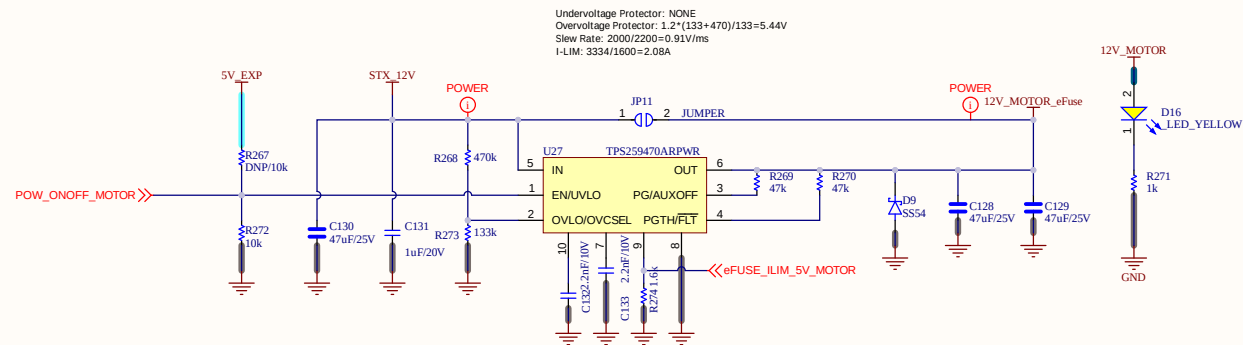
C

D

EFUSE for SLN & HD

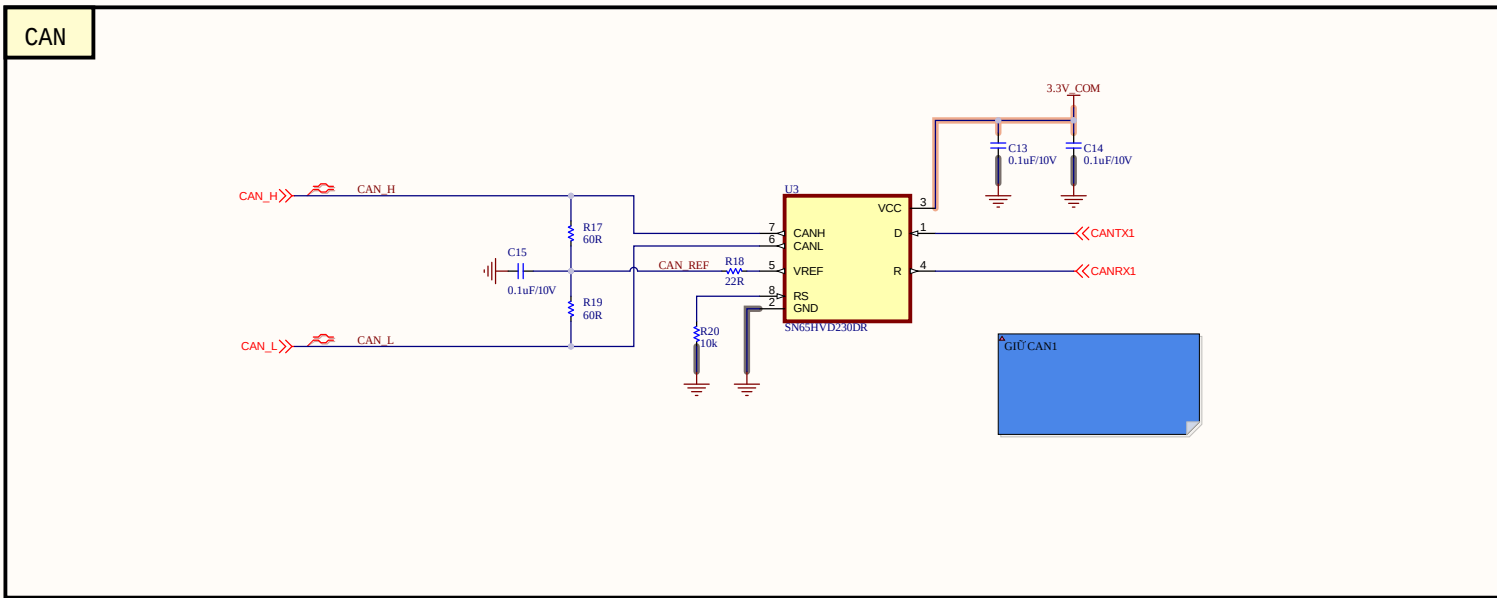


EFUSE for MOTOR



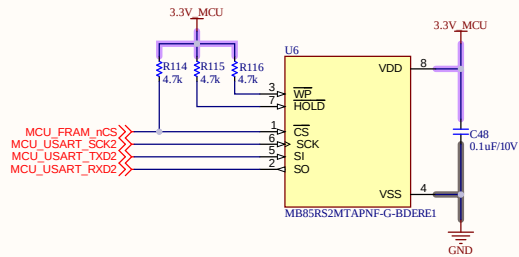
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Design by:	Duy Nguyen	Sheet title:	07_FUNCTION_EFUSE.SchDoc
Checked by:	Bao Bui	Doc no:	7
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	7 of 27
		Size:	A3
		Date:	11-Aug-26



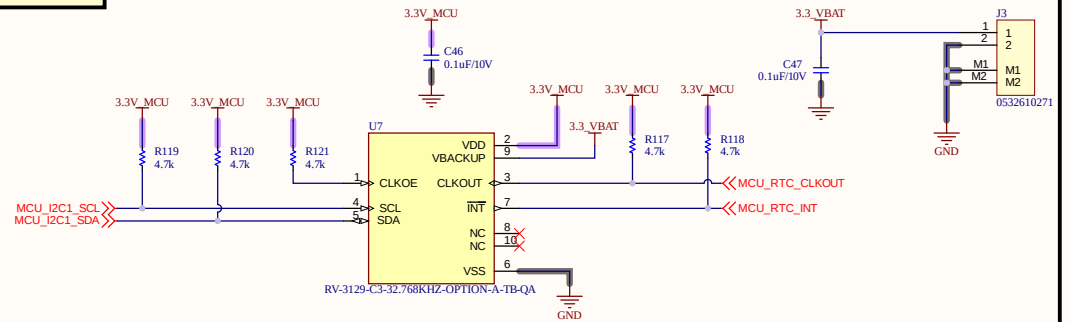


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Design by:	Duy Nguyen	Sheet title: 09_CAN.SchDoc	Sheet: 9 of 27
Checked by:	Bao Bui	Doc no: 9	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 11-Aug-26

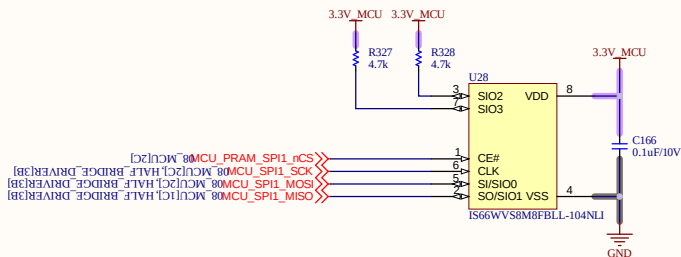
FRAM 2M



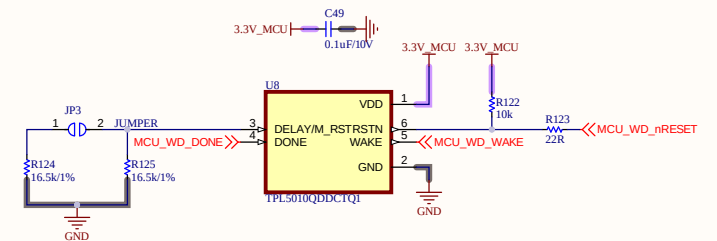
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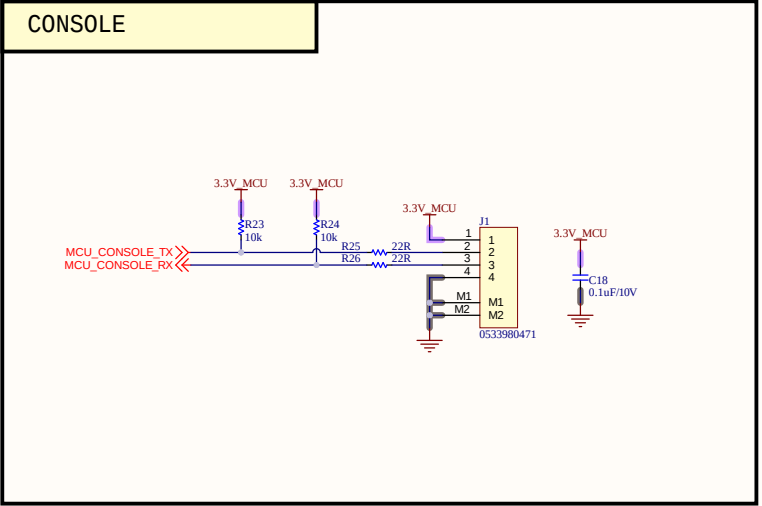
PRAM 64M



WATCH DOG

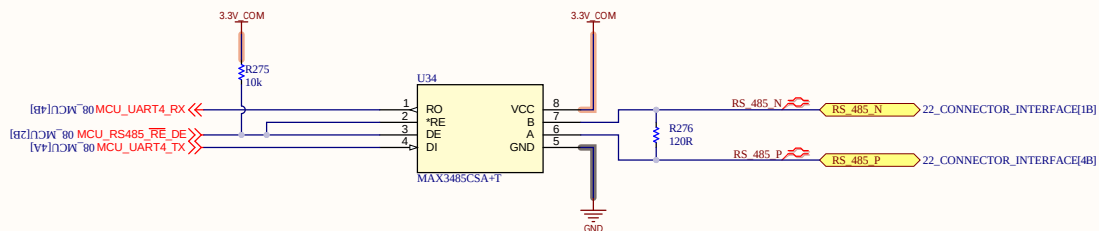


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Design by:	Duy Nguyen	Sheet title:	10_PRAM_RTC_WDT.SchDoc
Checked by:	Bao Bui	Doc no:	10
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	10 of 27
		Size:	A3
		Date:	11-Aug-26

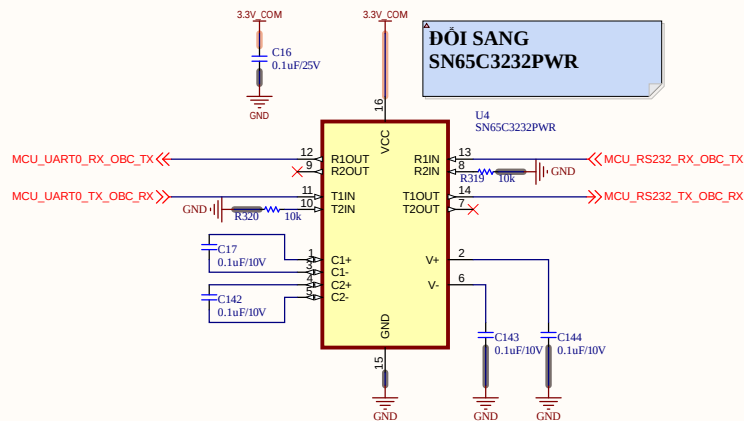


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Design by:	Duy Nguyen	Sheet title: 11_CONSOLE.SchDoc	Sheet: 11 of 27
Checked by:	Bao Bui	Doc no: 11	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 11-Aug-26

RS485

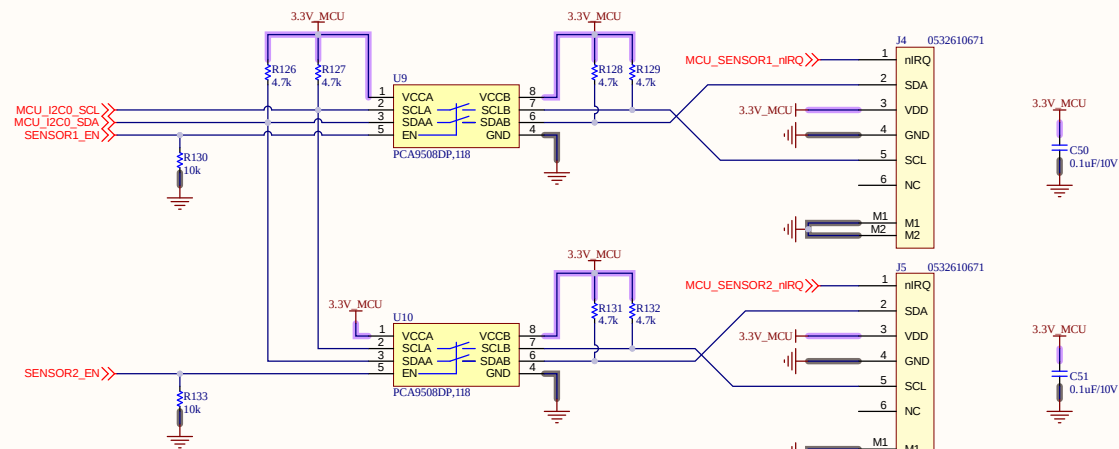


RS232



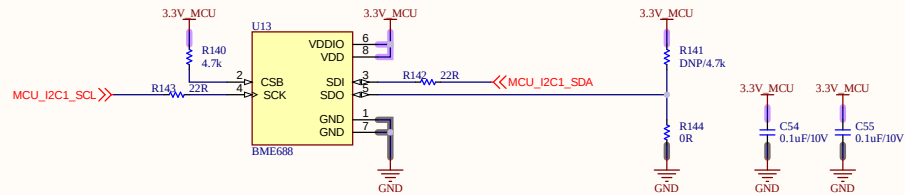
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Design by:	Duy Nguyen	Sheet title:	12_RS485_R232.SchDoc
Checked by:	Bao Bui	Doc no:	12
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	12 of 27
		Size:	A3
		Date:	11-Aug-26

I2C

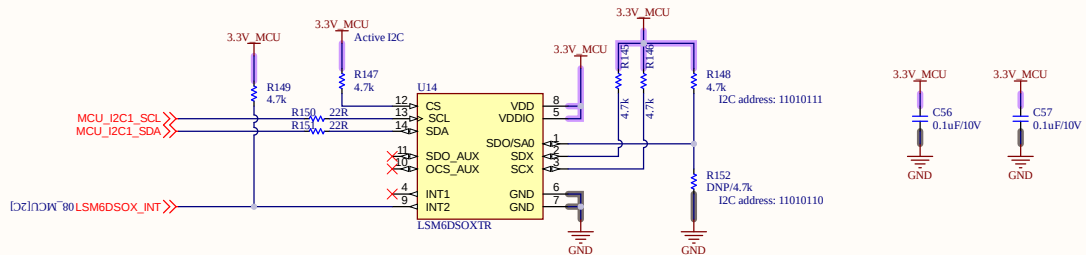


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Checked by:	Bao Bui	Doc no:	13
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	13 of 27
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		Date:	11-Aug-26

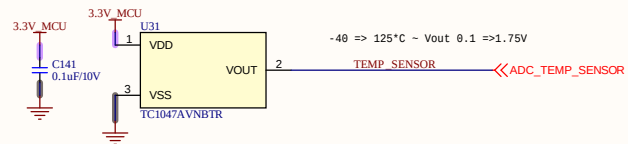
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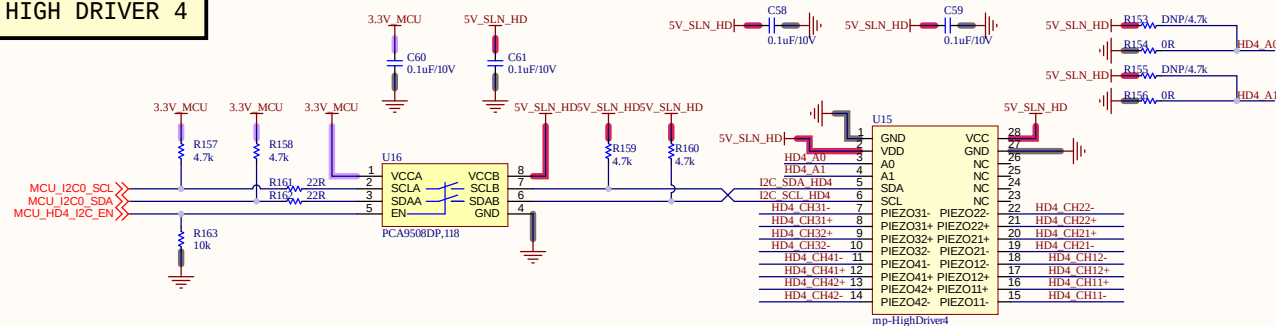


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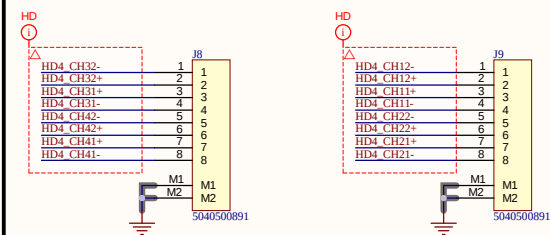


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Design by:	Duy Nguyen	Sheet title:	14_I2C_Sensor_OnBoard.SchDoc
Checked by:	Bao Bui	Doc no:	14
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	14 of 27
		Size:	A3
		Date:	11-Aug-26

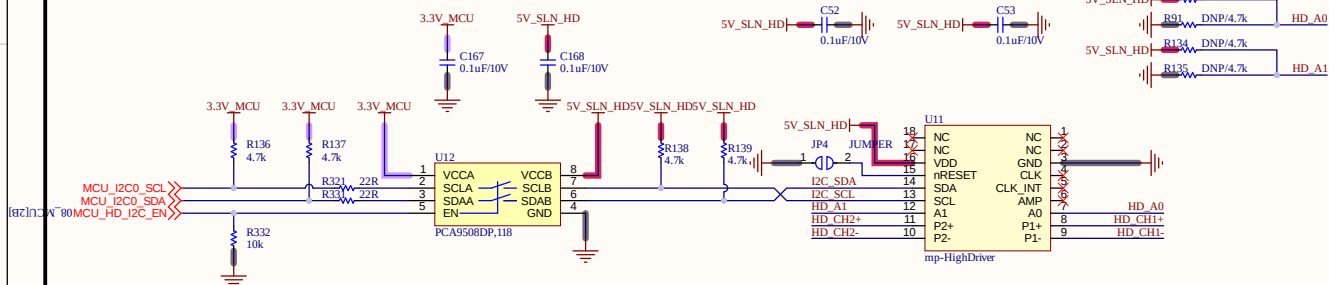
HIGH DRIVER 4



CONNECTOR BUMP

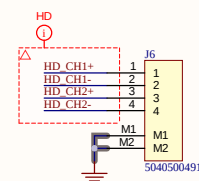


HIGH DRIVER



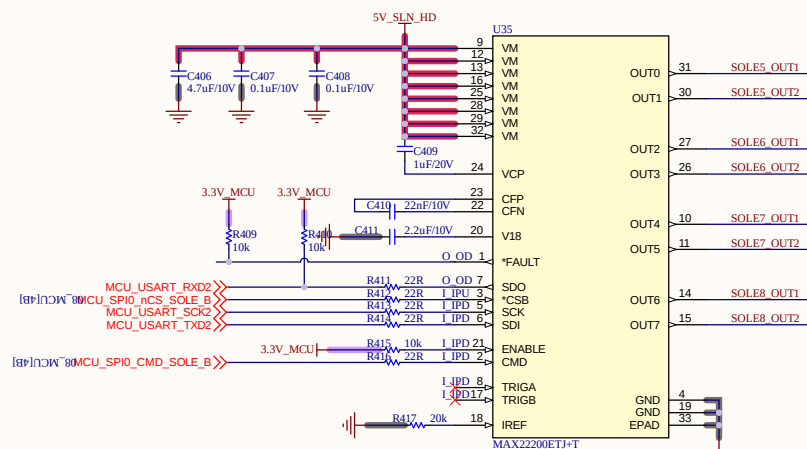
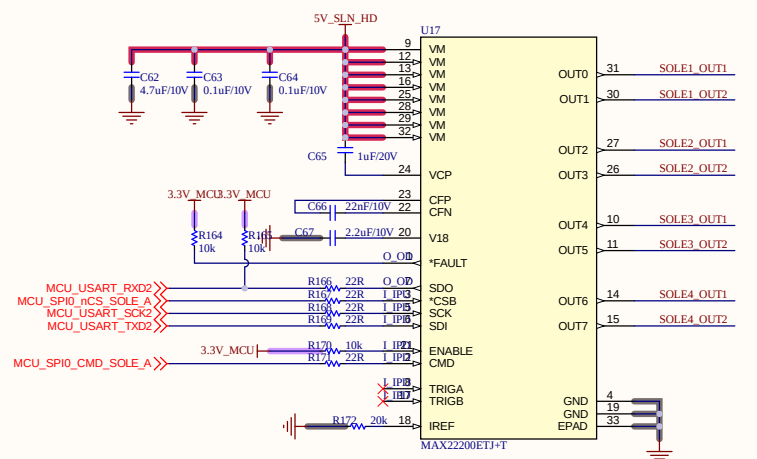
check pin lowdriver và highdriver

CONNECTOR BUMP

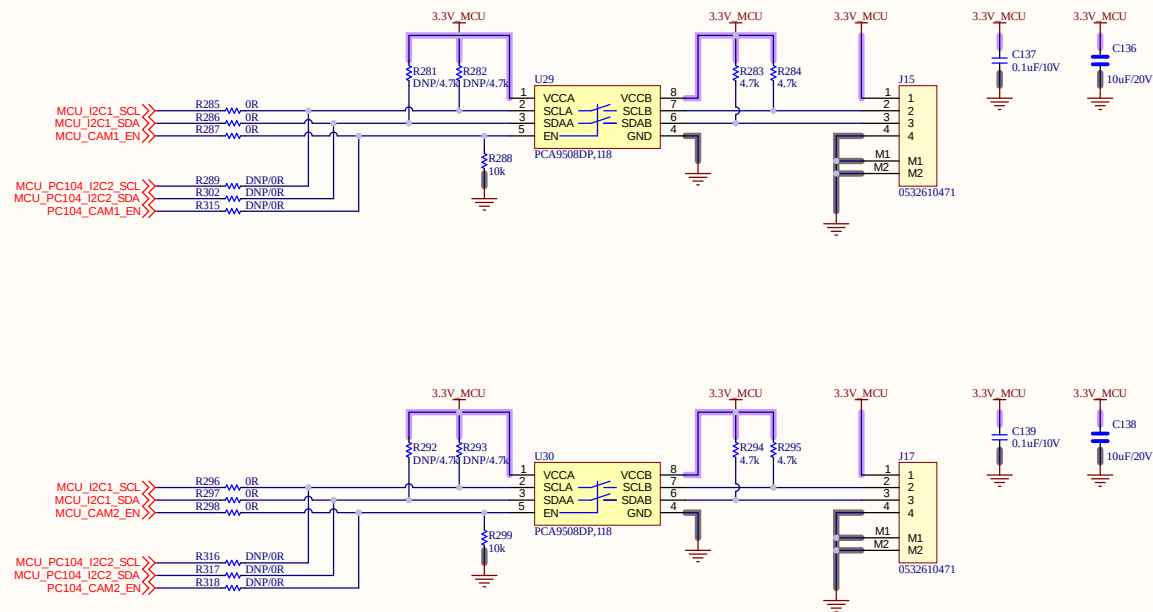


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Design by:	Duy Nguyen	Sheet title: 15_HD_PUMP.SchDoc	Sheet: 15 of 27
Checked by:	Bao Bui	Doc no: 15	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 11-Aug-26

SOLENOID

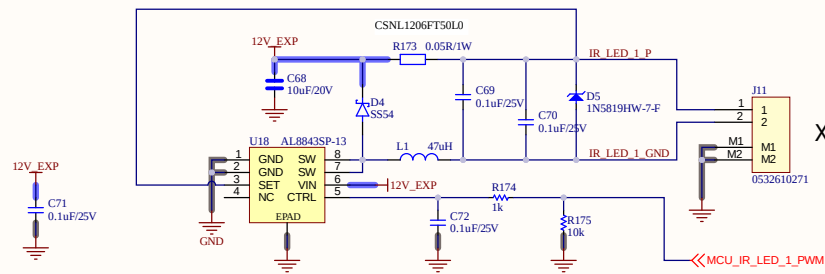


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Checked by:	Bao Bui	Doc no:	16
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	16 of 27
		Size:	A3
		Date:	11-Aug-26



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title: 17_CAM_CONNECTOR.SchDoc	Sheet: 17 of 27
Checked by:	Bao Bui	Doc no: 17	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 11-Aug-26

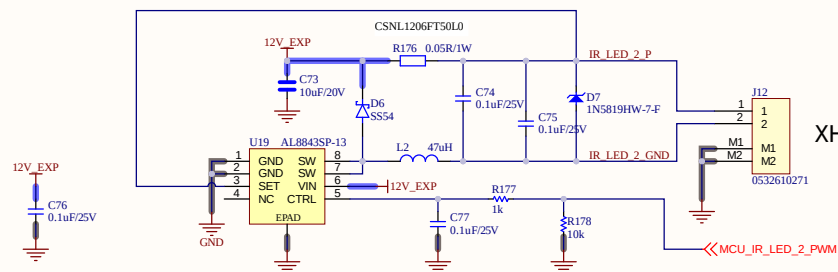
LED 1



XHP70D-H0-0000-0D0HN20E3

Đổi cuộn 3A
Tính lại trở

LED 2

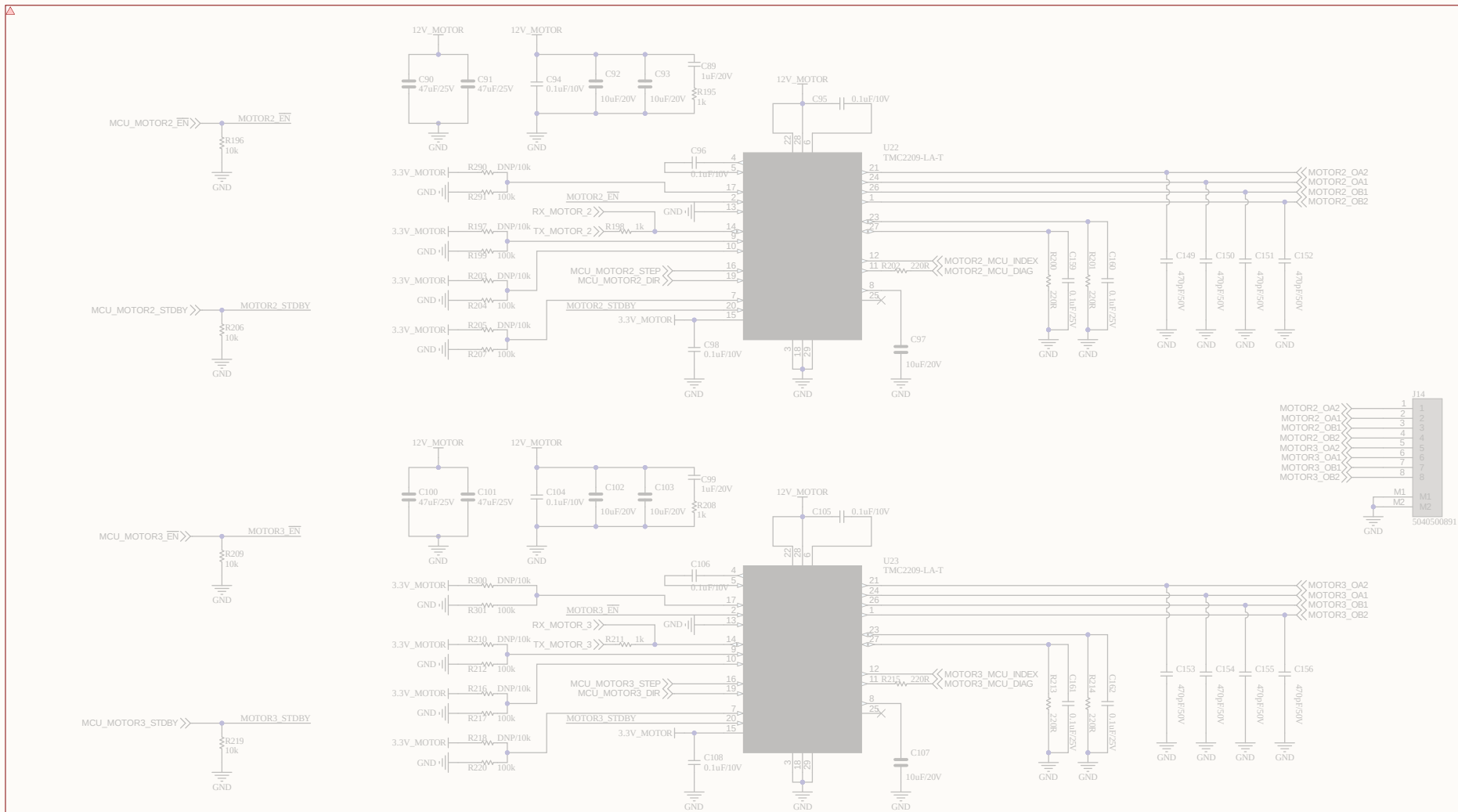


XHP70D-H0-0000-0D0HN20E3

Đổi cuộn 3A
Tính lại trở

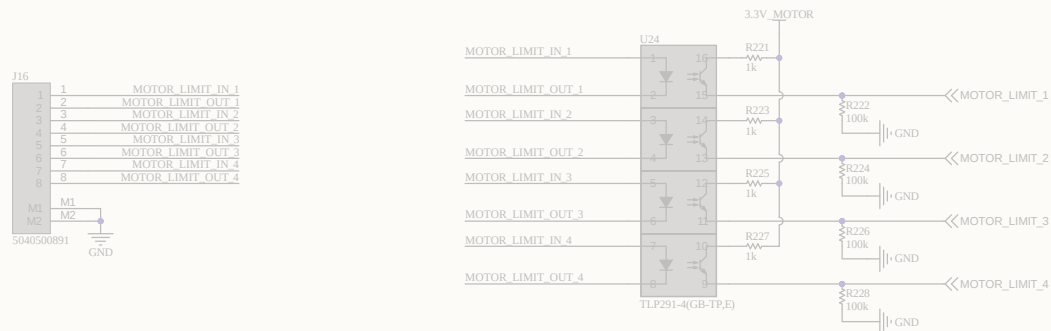


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Design by:	Duy Nguyen	Sheet title:	18_IR_LED.SchDoc
Checked by:	Bao Bui	Doc no:	18
Approved by:	Vu Pham	Revision	1.0.0
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		Size:	A3
		Date:	11-Aug-26

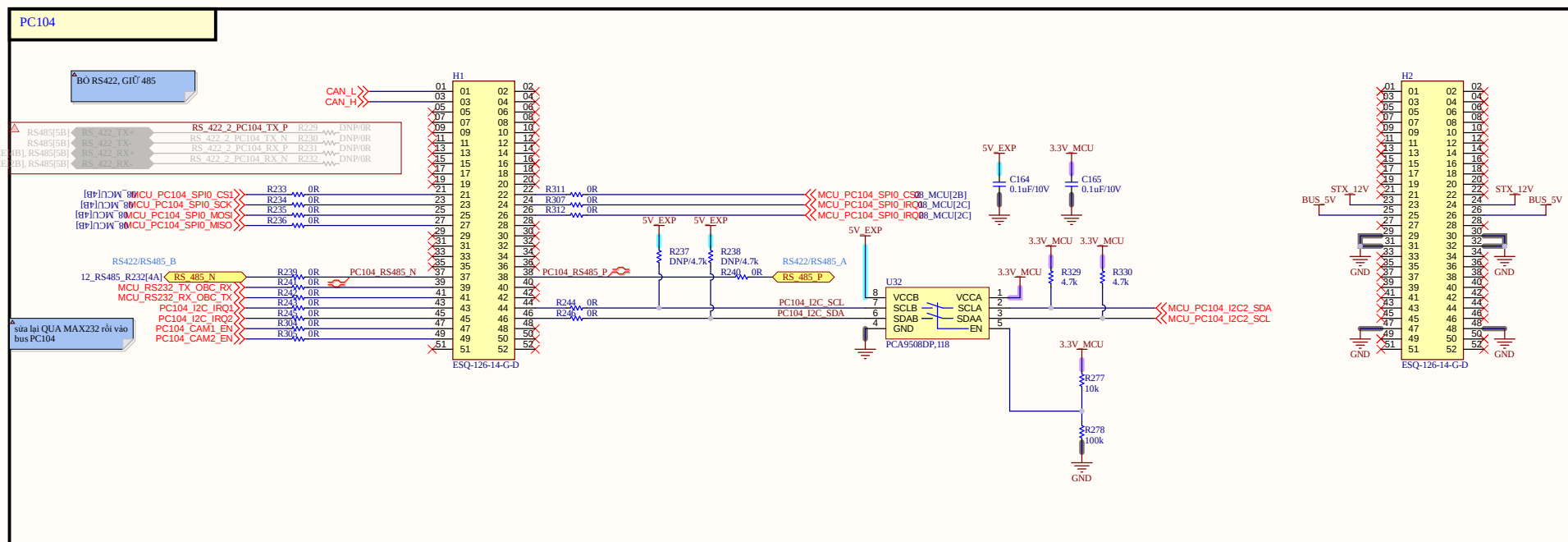


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Design by:	Duy Nguyen	Sheet title:	20_STEP_MOTOR_DRIVER_2.SchSheet
Checked by:	Bao Bui	Doc no:	20
Approved by:	Vu Pham	Revision	1.0.0
		Size:	A3
		Date:	11-Aug-26

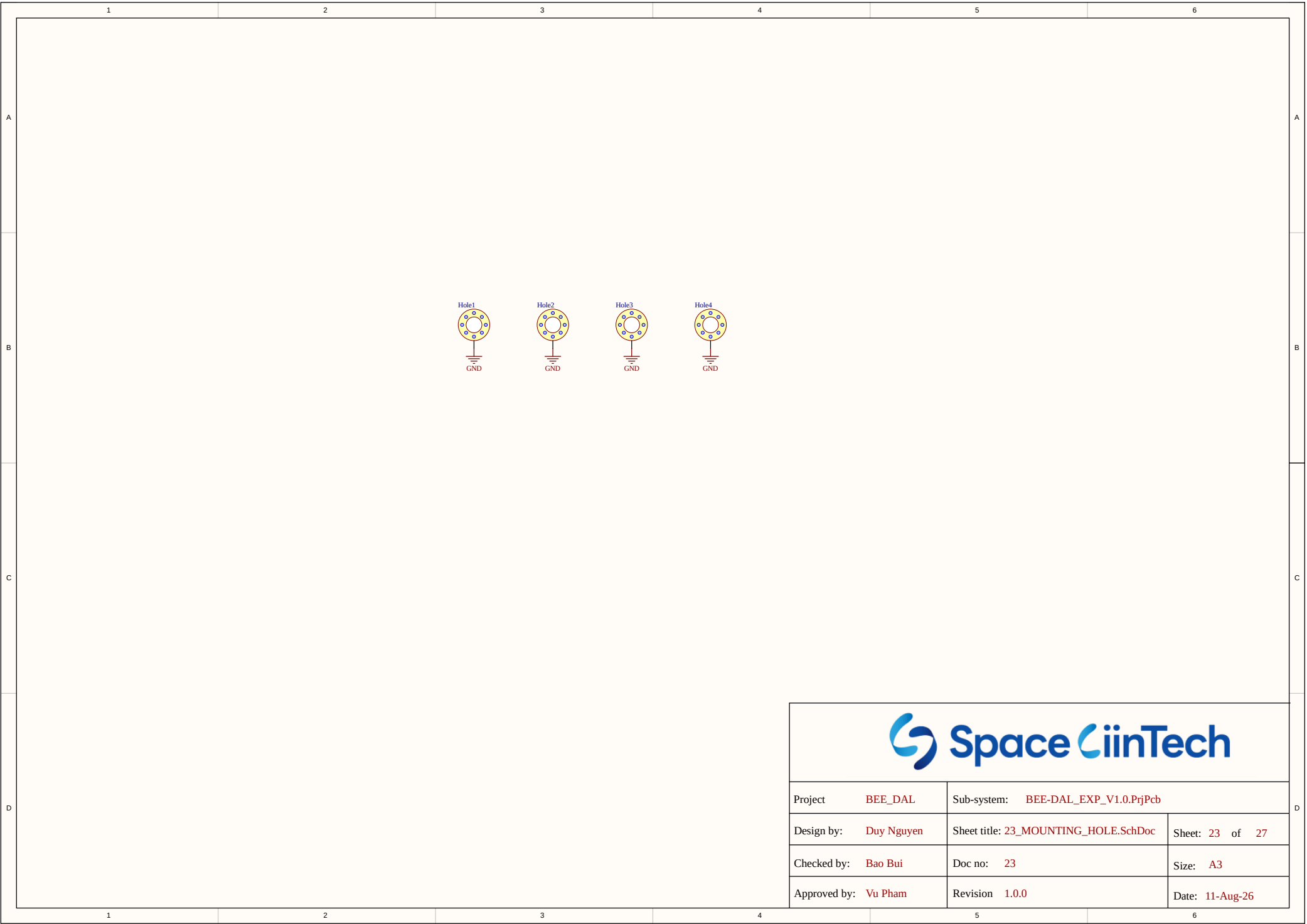
STEP LIMIT



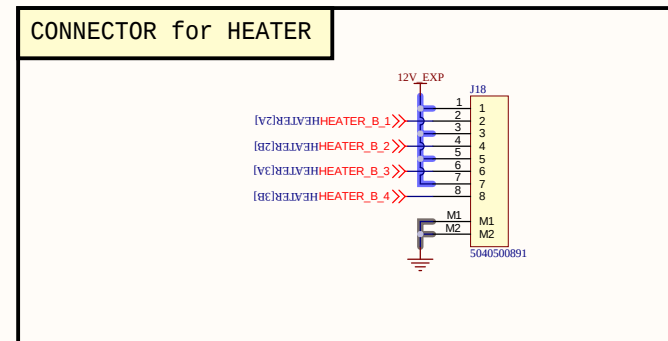
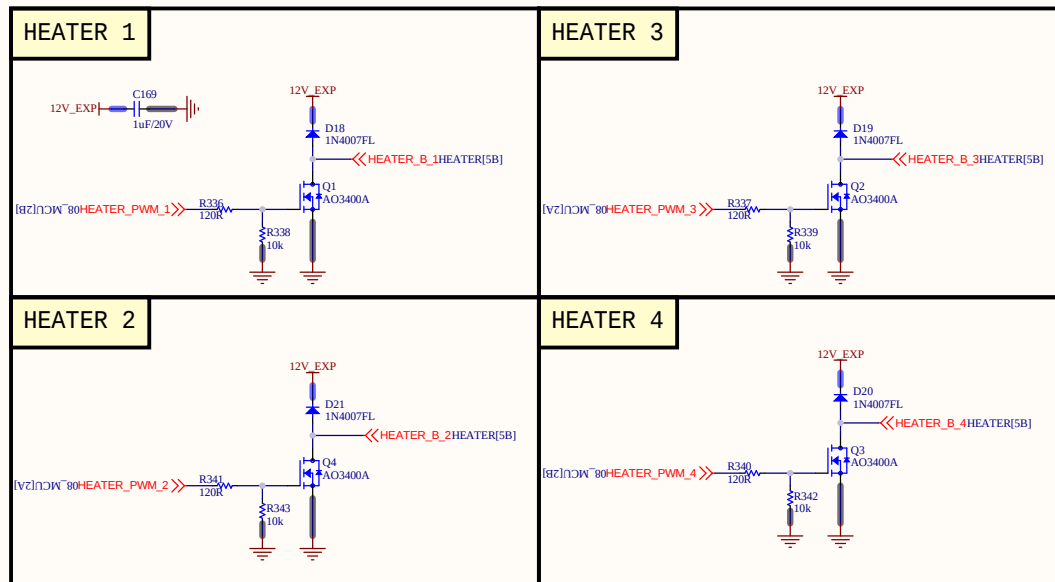
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Design by:	Duy Nguyen	Sheet title:	21_STEP_MOTOR_ISOLATOR_INOUT_SchDoc of 27
Checked by:	Bao Bui	Doc no:	21
Approved by:	Vu Pham	Revision	1.0.0
		Date:	11-Aug-26



Project	BEE_DAL	Sub-system: BEE-DAL_EXP_V1.0.PrjPcb	
Design by:	Duy Nguyen	Sheet title: 22_CONNECTOR_INTERFACE.SchDoc	Sheet: 22 of 27
Checked by:	Bao Bui	Doc no: 22	Size: A3
Approved by:	Vu Pham	Revision 1.0.0	Date: 11-Aug-26

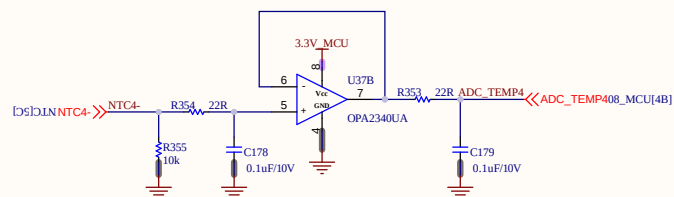
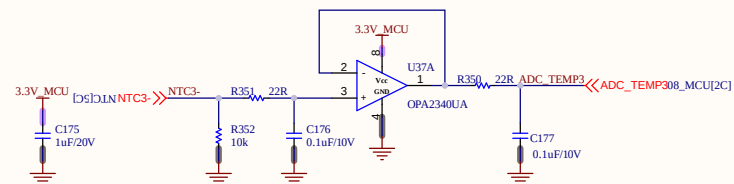
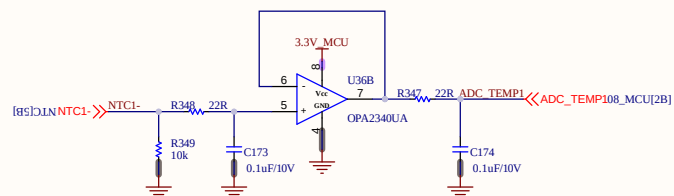
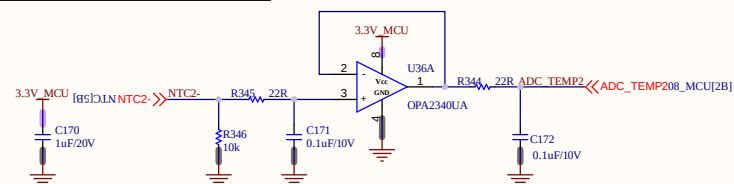


		
Project	BEE_DAL	Sub-system: BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title: 23_MOUNTING_HOLE.SchDoc
Checked by:	Bao Bui	Doc no: 23
Approved by:	Vu Pham	Revision 1.0.0
		Sheet: 23 of 27
		Date: 11-Aug-26

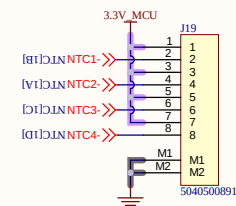


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Design by:	Duy Nguyen	Sheet title:	HEATER.SchDoc
Checked by:	Bao Bui	Doc no:	24
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	24 of 27
		Size:	A3
		Date:	11-Aug-26

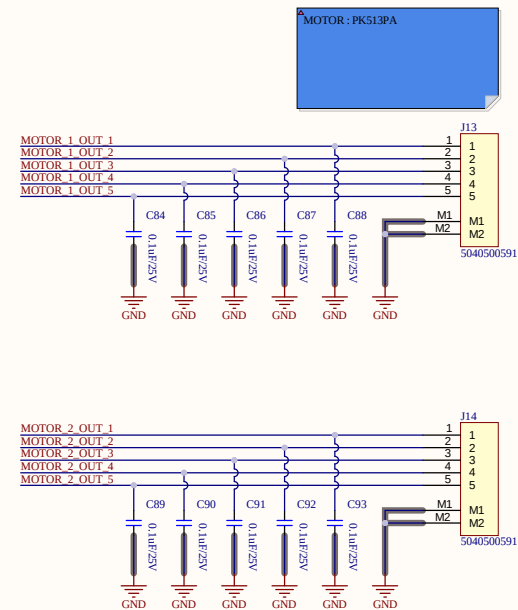
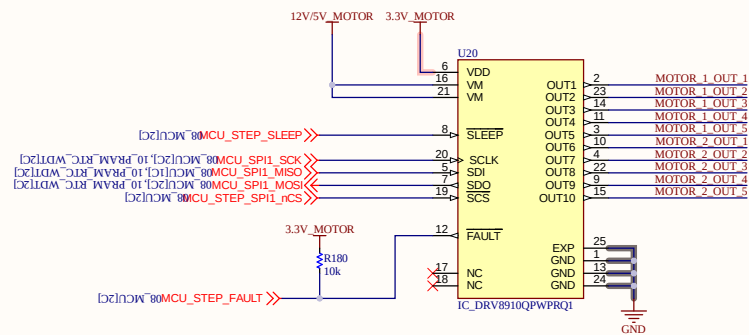
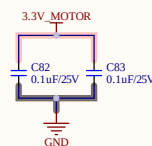
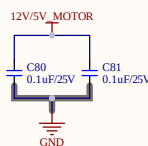
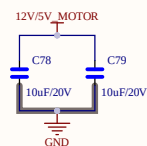
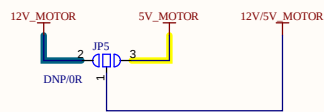
NTC



CONNECTOR FOR NTC

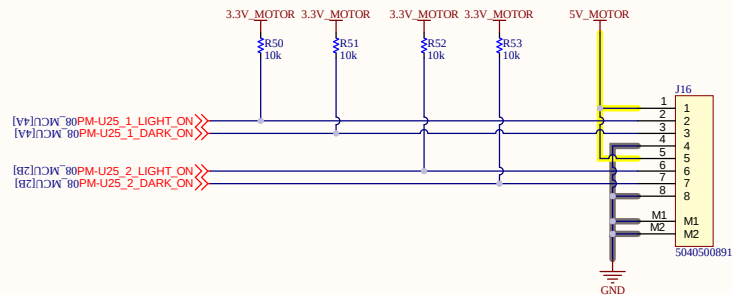


Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	NTC.SchDoc
Checked by:	Bao Bui	Doc no:	25
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	25 of 27
		Size:	A3
		Date:	11-Aug-26

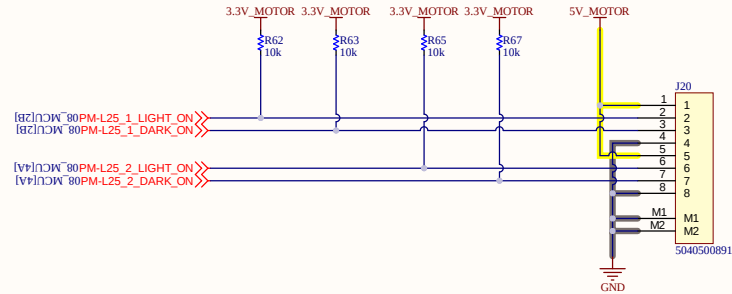


Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	HALF_BRIDGE_DRIVER.SchDoc
Checked by:	Bao Bui	Doc no:	26
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	26 of 27
		Size:	A3
		Date:	11-Aug-26

CONNECTOR for PM-U25



CONNECTOR for PM-L25



Project	BEE_DAL	Sub-system:	BEE-DAL_EXP_V1.0.PrjPcb
Design by:	Duy Nguyen	Sheet title:	LIMIT_SENSORS.SchDoc
Checked by:	Bao Bui	Doc no:	27
Approved by:	Vu Pham	Revision	1.0.0
		Sheet:	27 of 27
		Size:	A3
		Date:	11-Aug-26

